Testicular and Erectile Dysfunction

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Disclaimer

- Please note that the terms “male/female,” “men/women”, and “he/she” are used in most references and refer to **biological sex** and not gender identity
- The intersex community is not discussed in the scope of this session
- Couple infertility in this lecture refers specifically to **cisgender heterosexual couples** and does not include non-cisgender and non-heterosexual couples

Leaning objectives

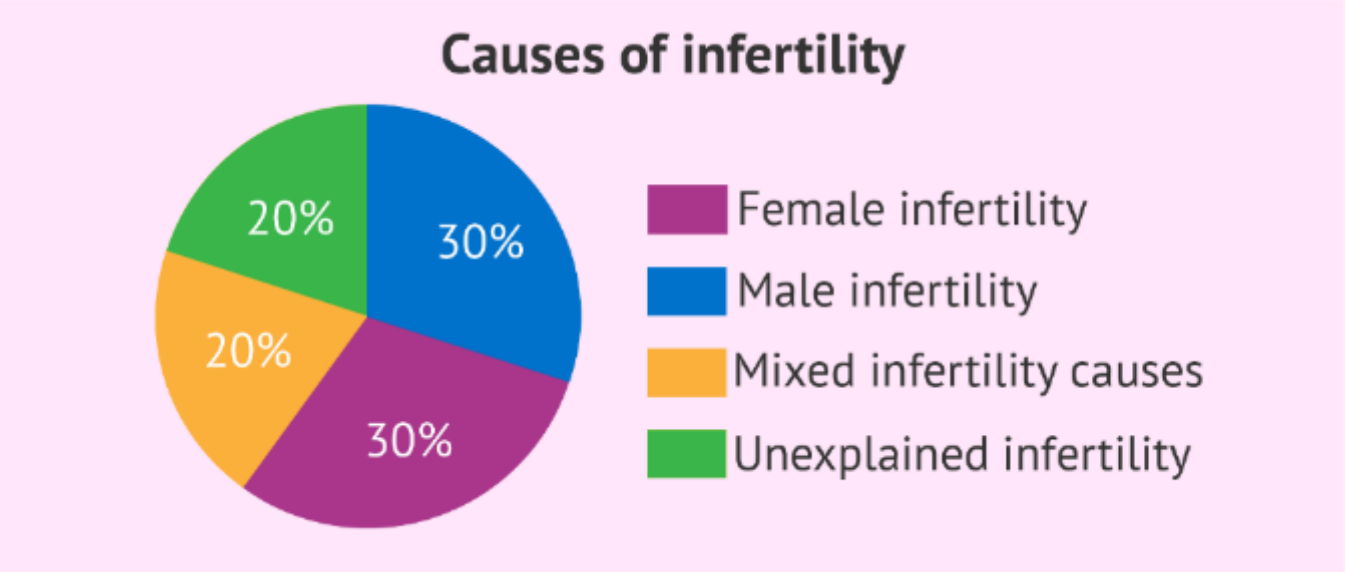
- Define infertility and recognize the contribution of male factors.
- Describe the medical history as well as medication and substance exposures that may impact male fertility.
- Describe the roles of physical exam, semen analysis, endocrine evaluation, and genetic workup in male infertility.
- Describe the medical and surgical managements of male infertility.
- Explain the 5 major types/causes of erectile dysfunction.
- Describe the evaluation and management of erectile dysfunction.

Infertility: Definition

- Primary infertility
  - Inability to achieve pregnancy after **12 months** of regular, unprotected intercourse (**6 months if female partner is aged ≥35 years**)
- Secondary infertility
  - As above, but in the setting of previously having at least one successful pregnancy and live birth

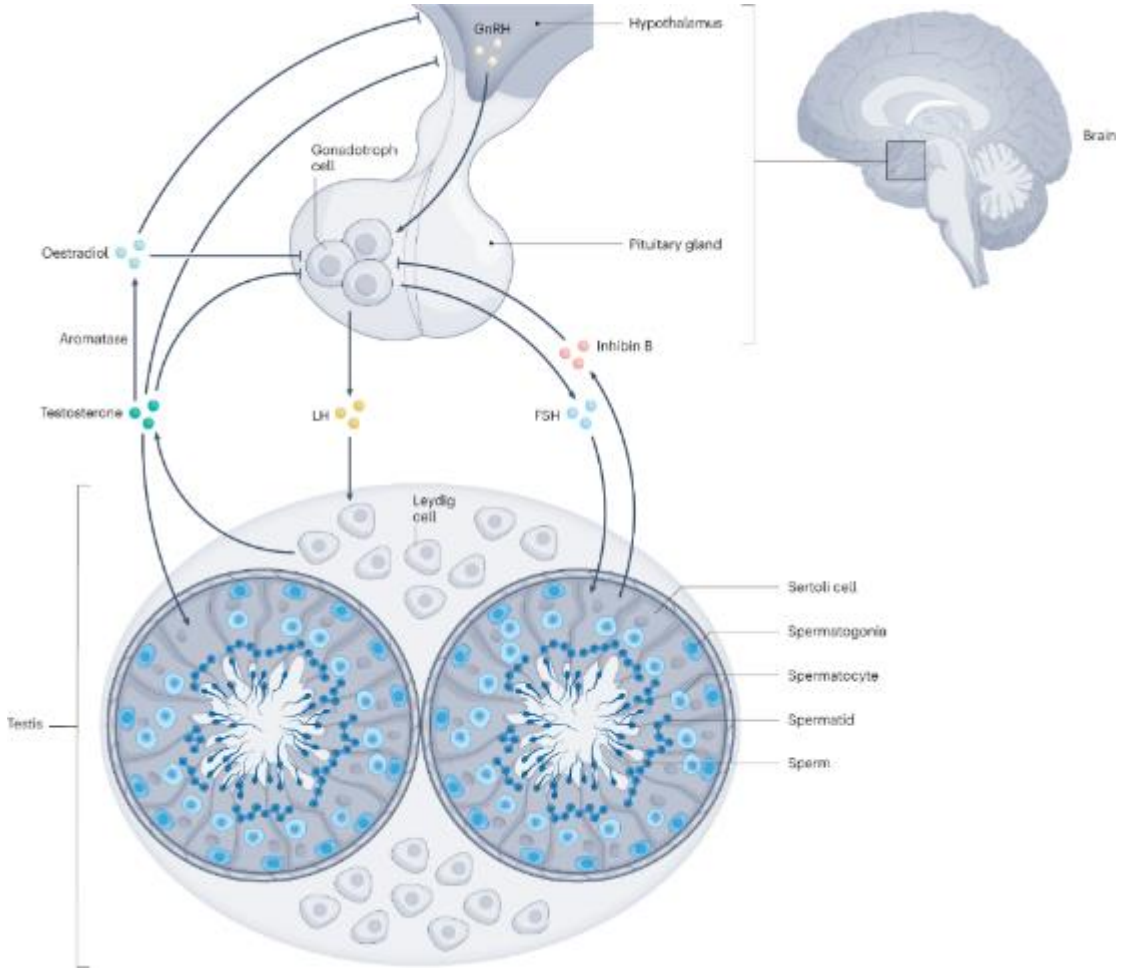
Infertility: Prevalence

- **15%** of all couples are infertile
- **50%** of infertility cases involve male factor
- **2%** of all infants born in the USA are conceived via IVF



https://www.invitra.com/en/sterility-causes/causes-of-sterility/

Hypothalamic-pituitary-gonadal (HPG) axis

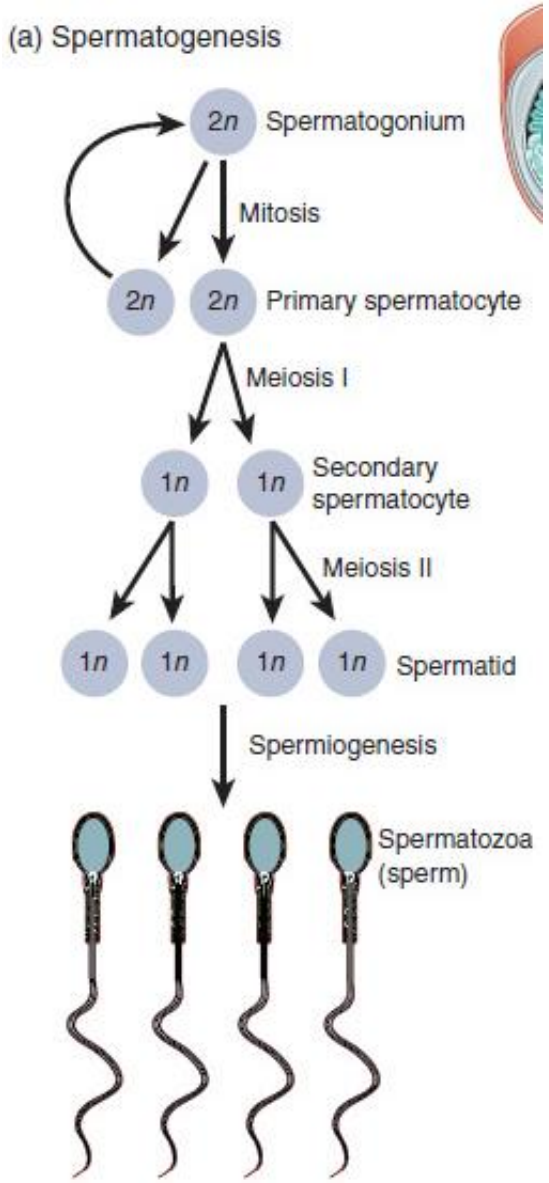


- Intratesticular testosterone (ITT)
  - Levels are >80 times higher than serum T
  - Critical for spermatogenesis
- Exogenous testosterone use
  - Causes suppression of HPG axis via negative feedback → undetectable FSH/LH → suppression of ITT → suppression of spermatogenesis → infertility & testicular atrophy
  - Up to 90% of men end up with **azoospermia** (no sperm in ejaculate) within 3 – 6 months

Eisenberg et al. Male infertility. *Nature Reviews Disease Primers*. 2023;9:49

Spermatogenesis

- Takes 64 – 74 days to complete
- Sperm transport through epididymis and vas deferens takes additional 14 days
- Clinical implication
  - Semen analysis to assess the outcome of lifestyle interventions (e.g. nicotine cessation), medical therapy (e.g. hormone supplementation, antibiotic therapy), or surgery (e.g. varicocelectomy) should be performed approximately **90 days (3 months)** later to realize the true effects of treatment



https://upload.wikimedia.org/wikipedia/commons/b/b1/Figure_28_01_04.jpg

Causes of male infertility

Pre-testicular	Testicular	Post-testicular	Behavioral/sexual dysfunction
1. Idiopathic hypogonadotropic hypogonadism **2. Kallman syndrome** 3. Pituitary disease or tumor (e.g. **prolactinoma**) 4. Pituitary radiation or surgery **5. Anabolic steroid use** 6. Androgen deprivation therapy (ADT) 7. Feminizing hormone therapy	**1. Klinefelter's syndrome (47,XXY)** **2. Varicocele** 3. Undescended testicle 4. Orchitis 5. Testicular torsion **6. Testicular cancer** 7. Environmental exposures 8. Testicular radiation 9. Chemotherapy 10. Orchiectomy **11. Febrile illness** **12. Excess heat**	1. STI, epididymitis, prostatitis **2. Congenital absence of the vas deferens (cystic fibrosis)** **3. Occlusion of vas deferens (due to vasectomy or injury from inguinal hernia repair)** 4. Ejaculatory duct obstruction **5. Prostate surgery** 6. Retroperitoneal surgery 7. Diabetic neuropathy	1. Inaccurate timing or technique of sexual intercourse 2. Use of lubricants (**nonoxynol-9**) **3. Erectile dysfunction** 4. Hypoactive sexual disorder 5. Decreased libido 6. Peyronie's disease 7. Hypospadias 8. Delayed ejaculation

Male infertility: Office evaluation

- History
  - Childhood development and surgical procedures
  - Infection history
  - Medical history with attention to endocrinopathy or delayed puberty
- Medication history (drugs affecting spermatogenesis, ejaculation, or testosterone production)
- **Review of female partner fertility and risk factors**
  - If concerns (e.g. irregular menses, advanced maternal age, recurrent pregnancy losses) → referral for evaluation by female infertility specialist (i.e. REI)
- Physical examination
  - Stigmata of hypogonadism (e.g. decreased hair distribution, bone and muscle mass, gynecoid pelvis)
  - Any inguinal or scrotal scars suggesting prior surgery
  - Inspection of penis for hypospadias or other abnormalities
- Scrotal examination
  - Assessment of testicular volume and consistency, identification of varicocele, presence of both vas deferens
  - Normal testicular volume: **15 – 20-cc**
  - Normal testicular length: **3.5 – 5.0-cm**
- Digital rectal exam may reveal prostatitis or ejaculatory duct obstruction (associated with dilatation of seminal vesicles)

Male infertility: Laboratory evaluation

- Semen analysis
  - If abnormal → repeat
  - Volume x concentration x motility = **total motile sperm count (TMSC)**
    - Most important parameter
    - Normal: ≥40 million
- Oligozoospermia: Low sperm concentration
- Asthenozoospermia: Low motility
- Teratozoospermia: Low % normal morphology
- **Azoospermia:** No sperm seen
  - Needs to be confirmed on **centrifuged specimen**
  - Obstructive (OA) vs. nonobstructive (NOA)

Table 1. WHO reference values for human semen

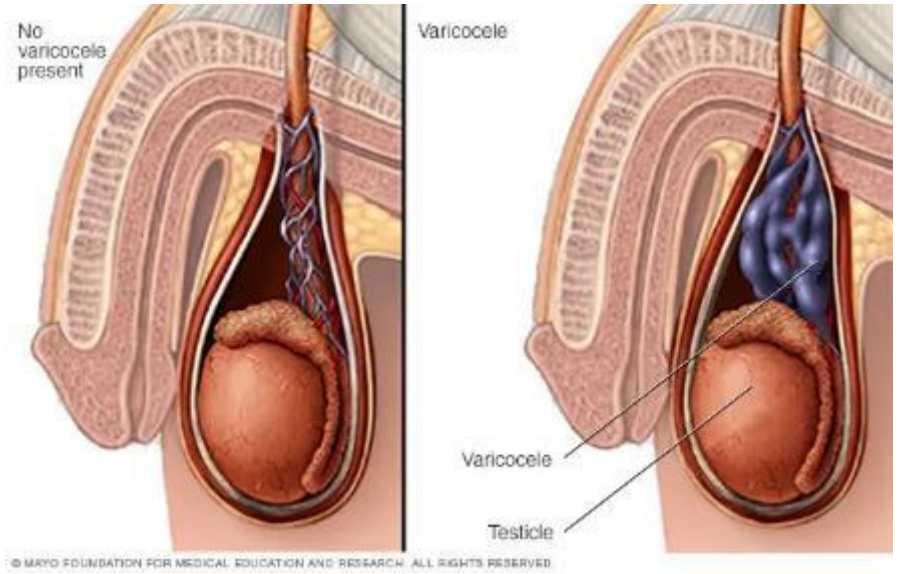
Semen Analysis Parameter	Normal Values
Sperm count Number of sperm in ejaculate	15 million/ml or more
Total motility Proportion of sperm that can swim	40% or more
Progressive motility Proportion of sperm swimming in one direction	32% or more
Morphology Size and shape of sperm	4% or more normal forms
Vitality Proportion of sperm that are alive	58% or more live sperm
Semen volume Amount of semen produced	1.5 ml or more
White blood cells Can indicate an infection	Less than 1 million/ml

Male infertility: Laboratory evaluation

- Serum endocrine testing (FSH, LH, testosterone)
  - If oligozoospermia, azoospermia, or signs of endocrinopathy (e.g. small testicles)
- Advanced sperm function tests (e.g. sperm DNA fragmentation index [DFI])
  - Elevated sperm DFI → infertility, increased risk of miscarriage, decreased ART success rates
- Genetic tests
  - If concentration <5 million/ml → **karyotype** (e.g. Klinefelter syndrome [47,XXY], Robertsonian translocation)
  - If concentration <1 million/ml → **Y chromosome microdeletion test**
    - AZFa/AZFb deletion: Incompatible with spermatogenesis
    - AZFc deletion: Spermatogenesis is still possible in some patients
  - If absent vas deferens → **CFTR mutation test + renal ultrasound**
    - Absent vas is associated with renal anomalies (e.g. renal agenesis, horseshoe kidney) due to common embryologic origin

Varicocele

- Abnormal dilation and tortuosity of the veins within the pampiniform plexus of the spermatic cord
- Present in
  - 15% of general population
  - 40% of primary male infertility
  - 80% of secondary male infertility
- **Most common reversible cause of male infertility**
  - Can also cause chronic testicular pain
- **85% are left sided only**
  - L testicular vein inserts into L renal vein at right angle
- Grading system
  - Grade 1: Only palpable with Valsalva maneuver
  - Grade 2: Palpable with and without Valsalva maneuver
  - Grade 3: Veins are visible through the skin (**"bag of worms"** appearance)



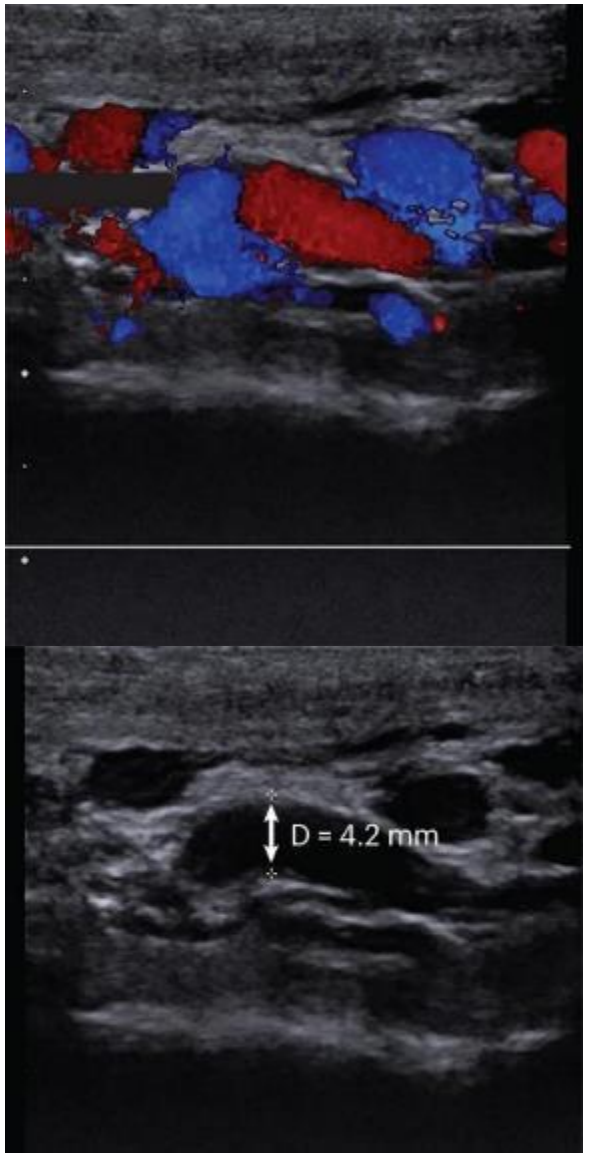


. 1. Example of a clinical grade III varicocele

https://www.mayoclinic.org/diseases-conditions/varicocele/symptoms-causes/syc-20378771

Varicocele: Diagnosis

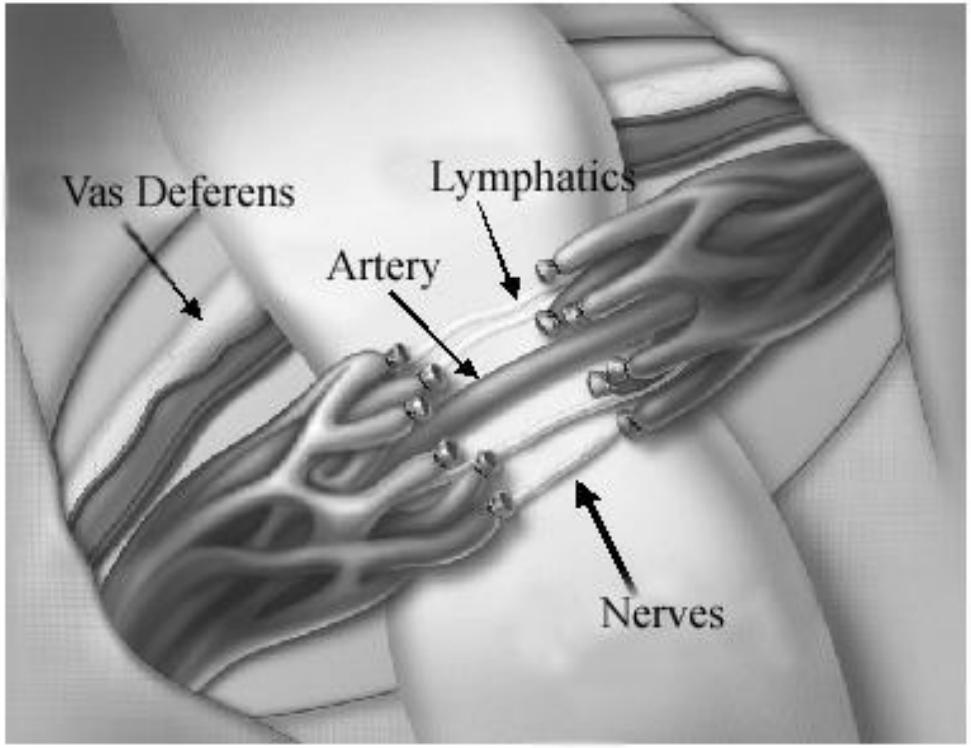
- Clinical diagnosis based on physical exam
- Scrotal ultrasound is generally not indicated
  - Varicocele that is seen on ultrasound but not palpated on exam (**“subclinical” varicocele**) is generally considered to be clinically insignificant
  - In certain patients who pose difficult exam (e.g. due to body habitus, robust cremasteric reflex), ultrasound may be helpful
- Varicocele is present if pampiniform plexus vein diameter is **>2.5 - 3.0-mm**, and the veins demonstrate **retrograde flow (venous reflux)** with Valsalva maneuver



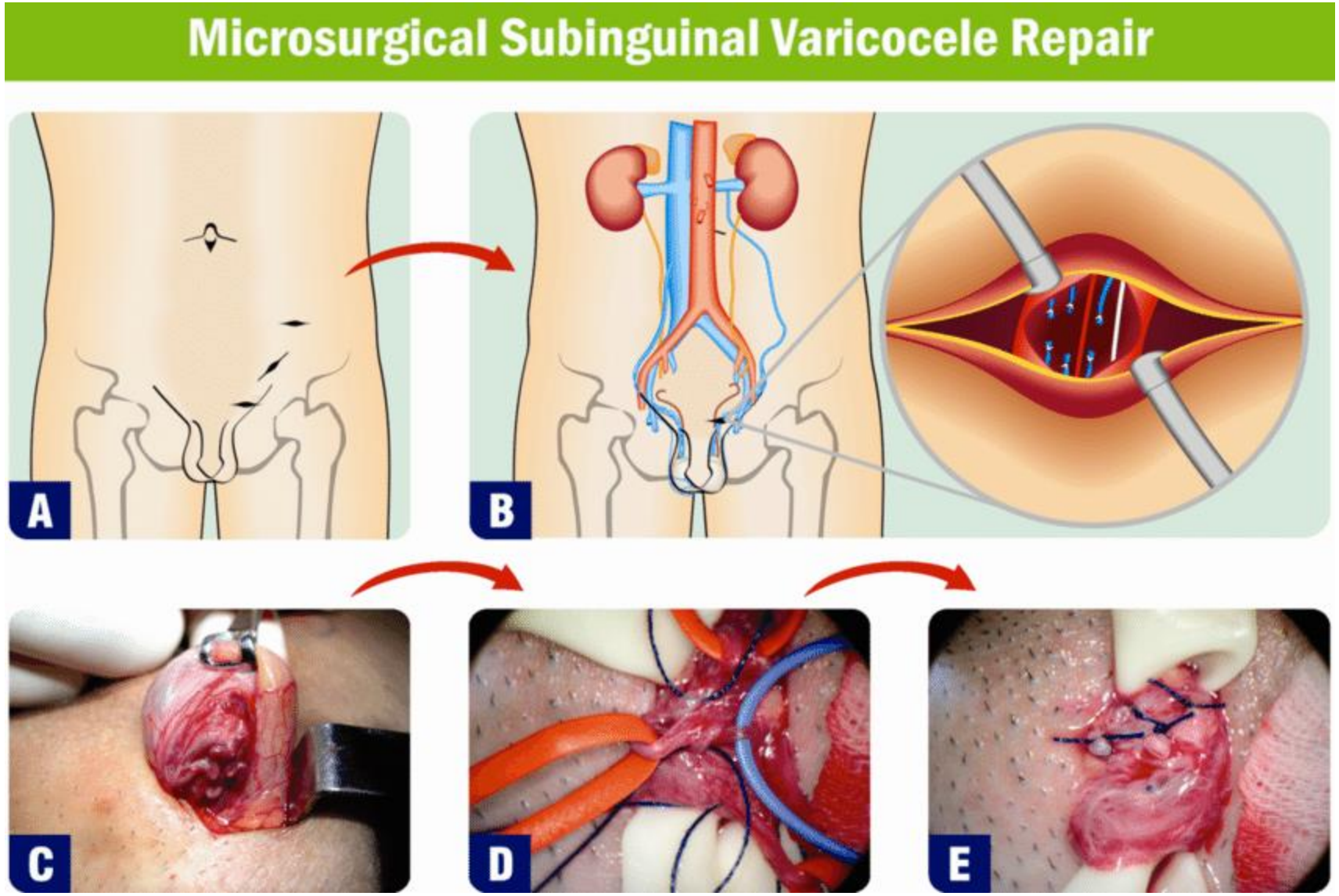
Eisenberg et al. Male infertility. *Nature Reviews Disease Primers*. 2023;9:49

Varicocele: Treatment

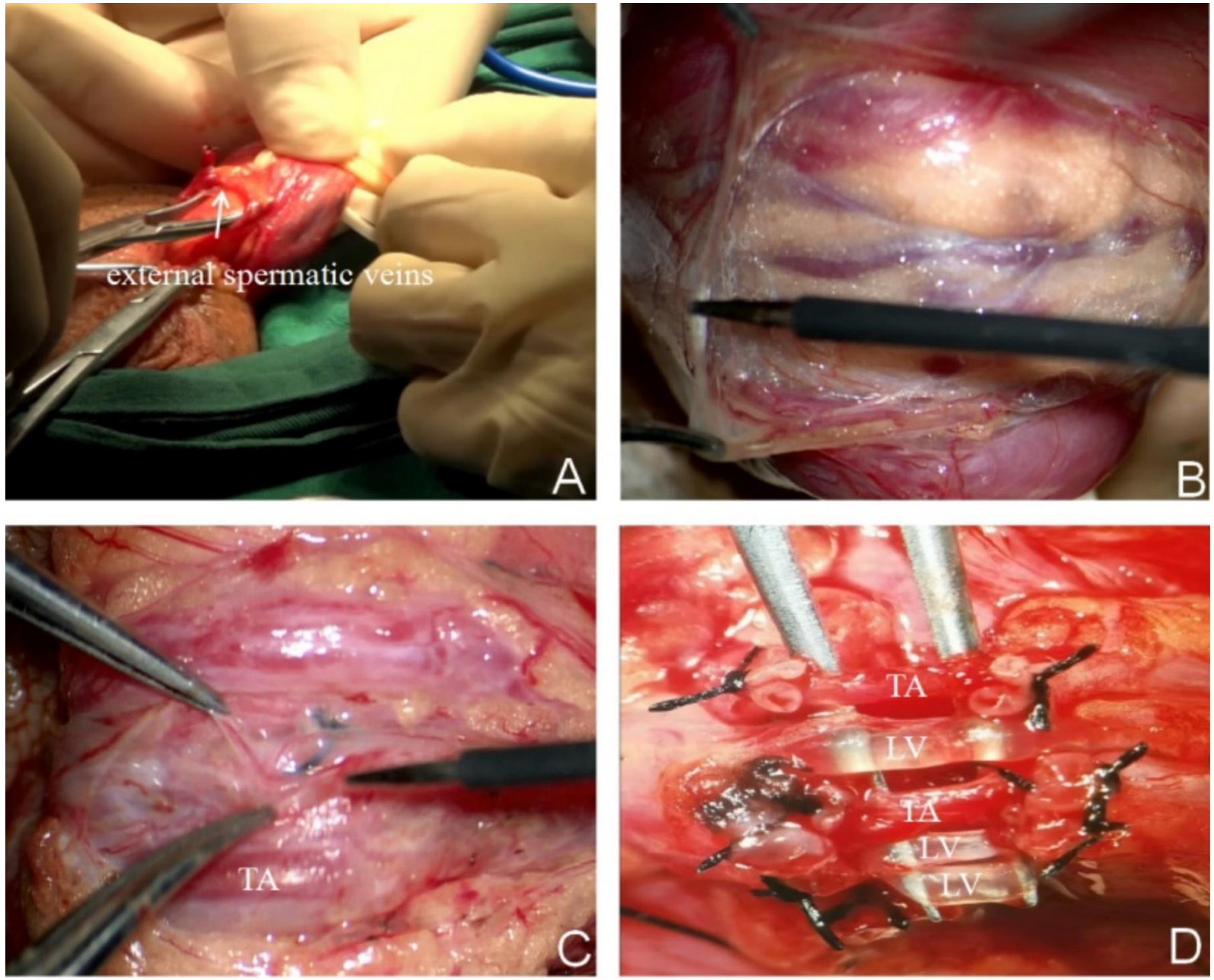
- Indications
  - Abnormal semen analysis
  - Chronic testicular pain
  - Testicular size discrepancy
- Surgical varicocelectomy
  - Microsurgical (gold standard)
    - Lowest rates of complications (hydrocele, testicular atrophy) and recurrence (≤1%)
  - Laparoscopic
    - Higher rates of complication and recurrence (up to 15%)
- Percutaneous angioembolization
  - Higher failure rate (20%) and recurrence rate (up to 13%)
- 70% of patients experience significant improvement in sperm parameters
  - Need to wait 3 – 6 months to see improvement
  - May also improve success rates of IVF



https://maleinfertility.org/procedures/microsurgical-varicocelectomy/surgical-treatment



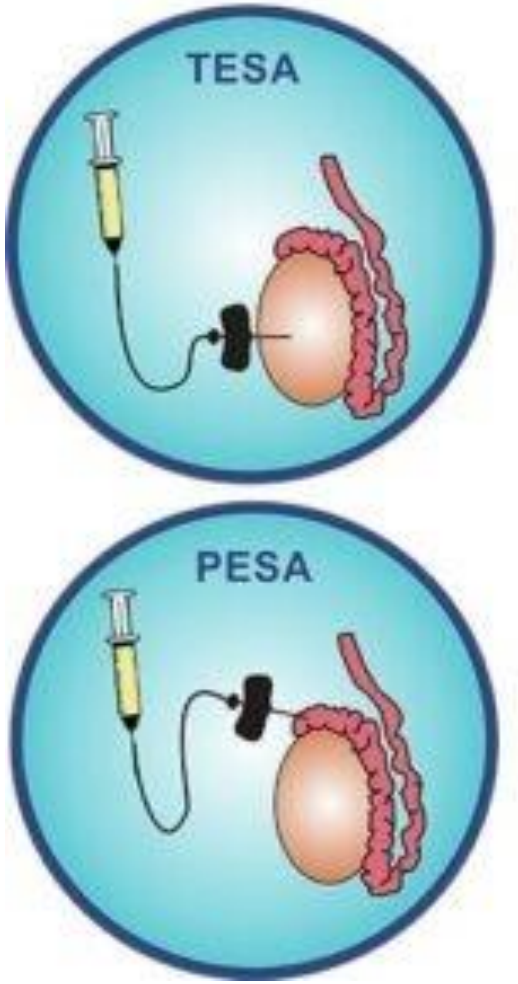
Esteves et al. *The Open Reproductive Science Journal*. 2011;3:16-26



Cao et al. Male infertility. *BMC Urology*. 2024;24:277

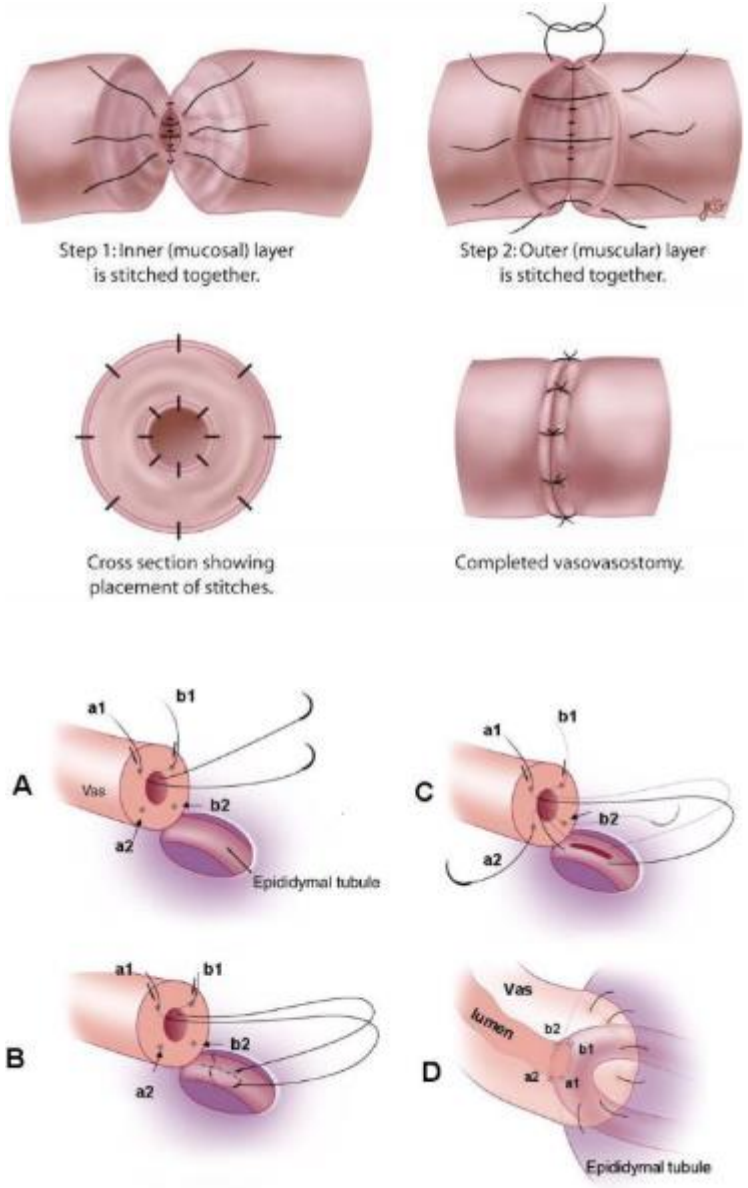
Patients with prior vasectomy

- >500,000 vasectomies/year
- Up to 12% have regret (e.g. due to new relationship)
- Options
  - Testicular sperm aspiration (TESA) or percutaneous epididymal sperm aspiration (PESA) with IVF
    - Also performed for other causes of **obstructive azoospermia** (e.g. cystic fibrosis)



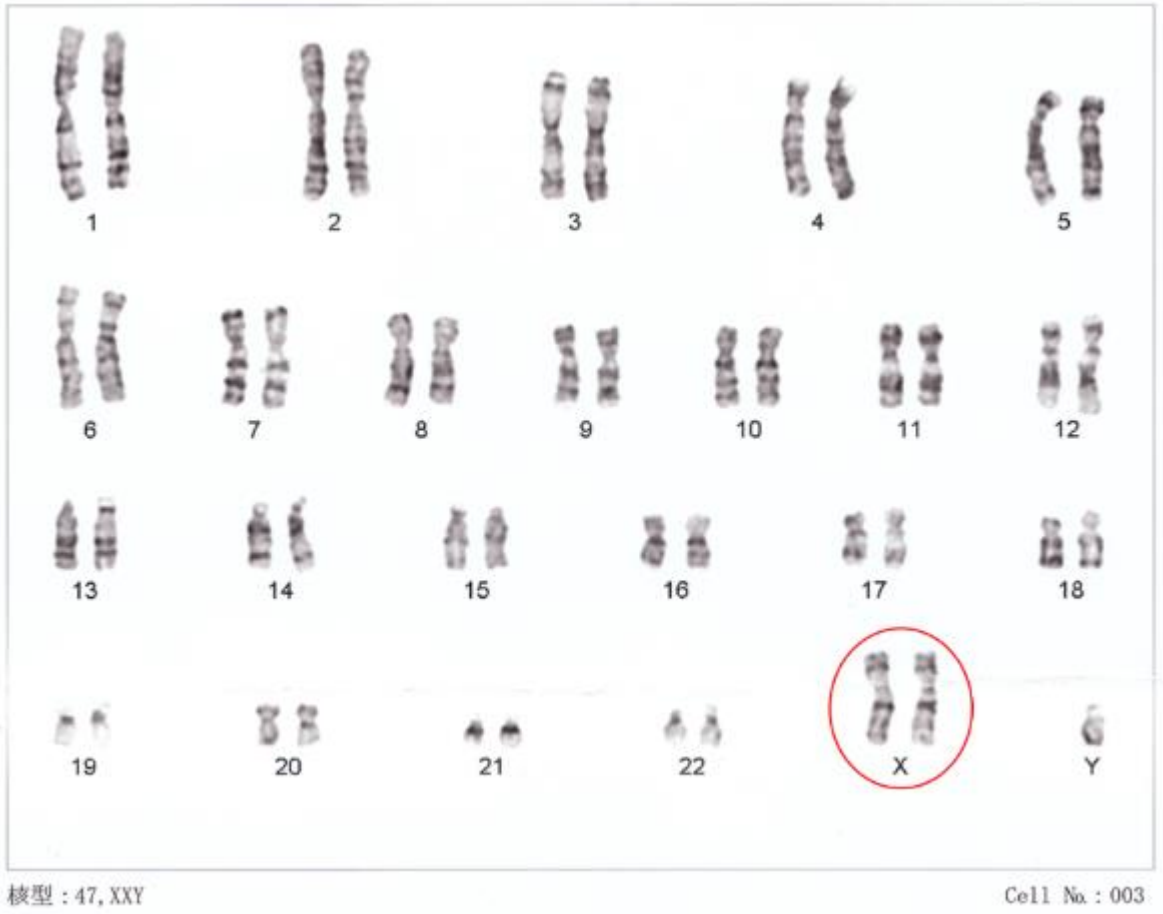
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- Options
  - Testicular sperm aspiration (TESA) or percutaneous epididymal sperm aspiration (PESA) with IVF
    - Also performed for other causes of **obstructive azoospermia** (e.g. cystic fibrosis)
  - Vasectomy reversal
    - Vasovasostomy
    - Epididymovasostomy



Klinefelter syndrome (KS)

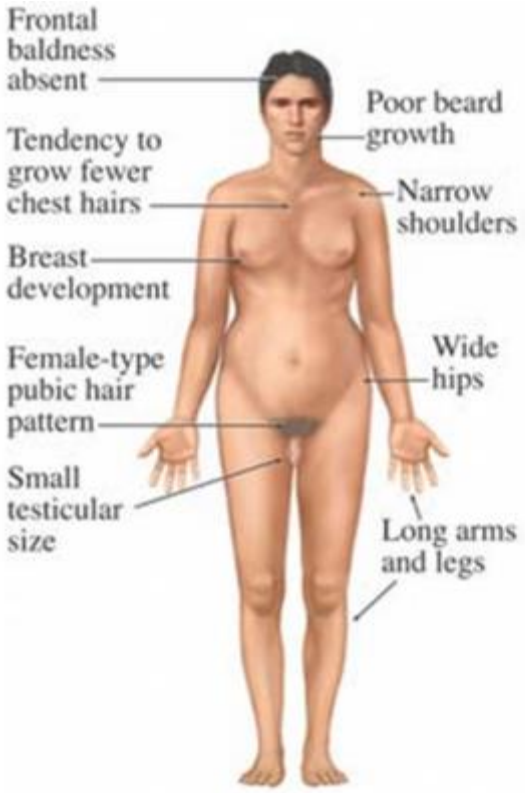
- Characterized by ≥2 X chromosomes and 1 Y chromosome
  - Most common: **47,XXY** (80 – 90%)
  - 48,XXXY and 49,XXXXY also described
- **Most common chromosomal disorder in males**
  - Incidence: 1 in 500 – 600
  - 3% of infertile men
    - 15% of azoospermic men
- Not inherited
- Advanced maternal age is only known risk factor
- Diagnosis: karyotype



https://en.wikipedia.org/wiki/Klinefelter_syndrome#/media/File:Human_chromosomesXXY01.png

KS: Clinical presentation

- Reading and speech difficulties (>75%)
- Autism (10%)
- Tall stature (30%)
- Gynecomastia (38 – 75%)
- Decreased facial/pubic hair (30 – 80%)
- **Small testes <6-cc (>95%)**
- **Nonobstructive azoospermia (>90%)**
- **Testosterone deficiency (63 – 85%)**
- Most patients are indistinguishable from 46,XY males
  - Classic physical exam findings are only seen in **30%** of patients
  - Only **25%** of KS patients are diagnosed (**<10%** before puberty)
  - Most commonly diagnosed from infertility work up





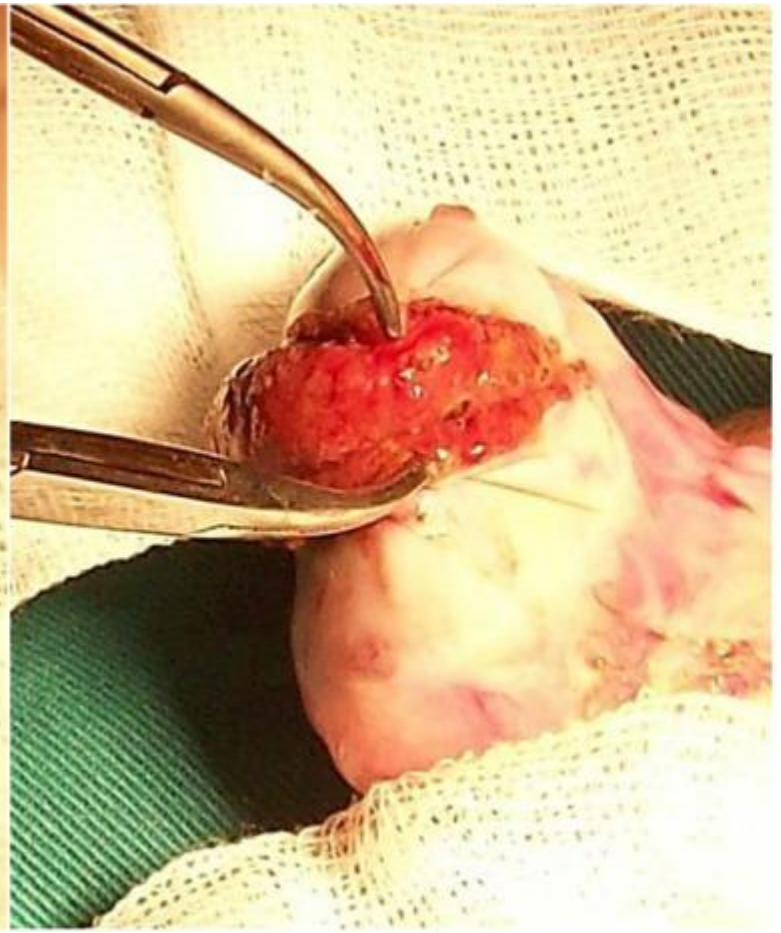
https://froemkelab.med.nyu.edu/surgery/content?ChunkIID=11722
https://en.wikipedia.org/wiki/Klinefelter_syndrome

KS: Fertility management

- If sperm in ejaculate (8%) → intrauterine insemination (IUI), *in vitro* fertilization (IVF)
- If no sperm in ejaculate (92%) → **microsurgical testicular sperm extraction (microTESE)** → intracytoplasmic sperm injection (ICSI)

MicroTESE

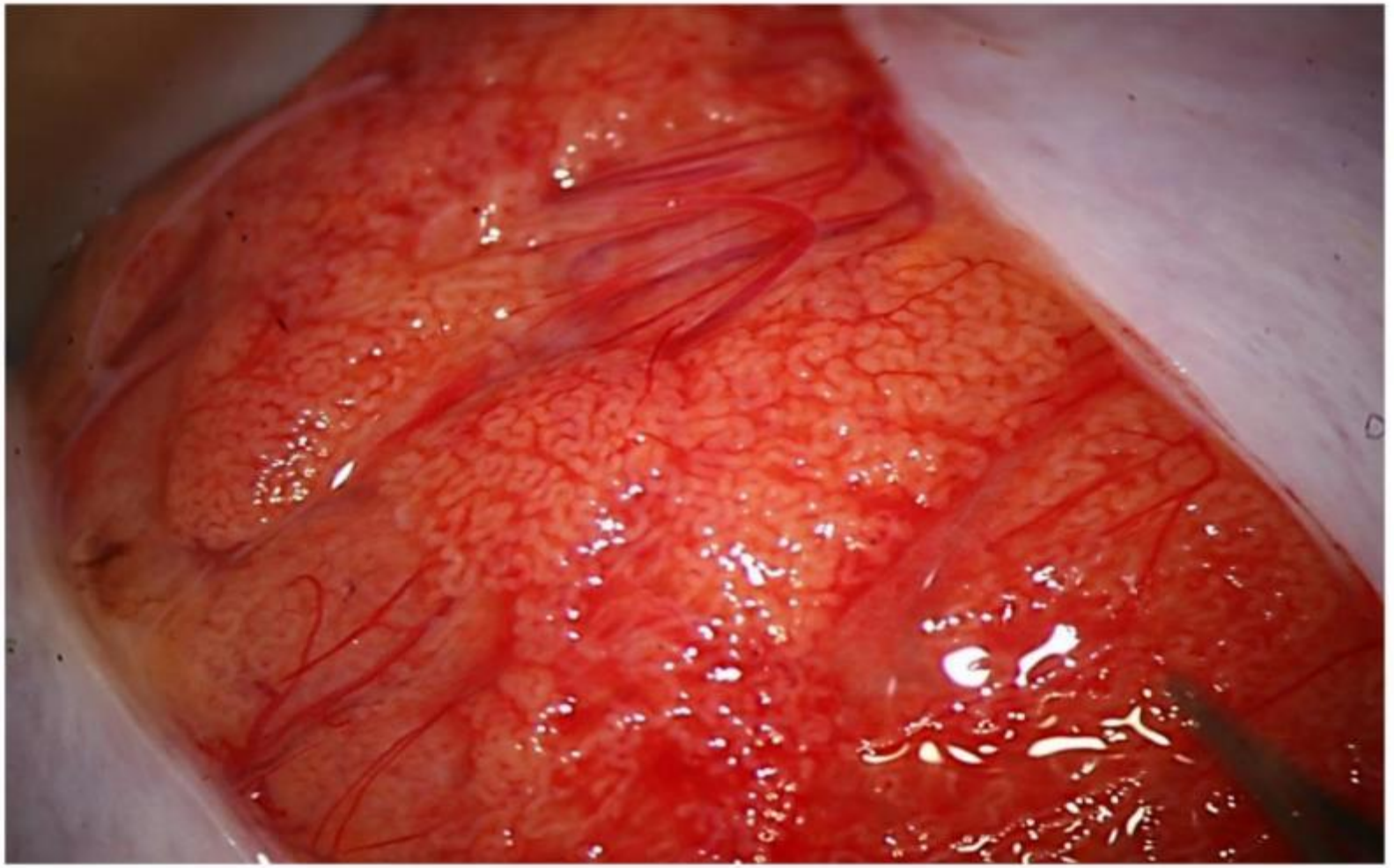






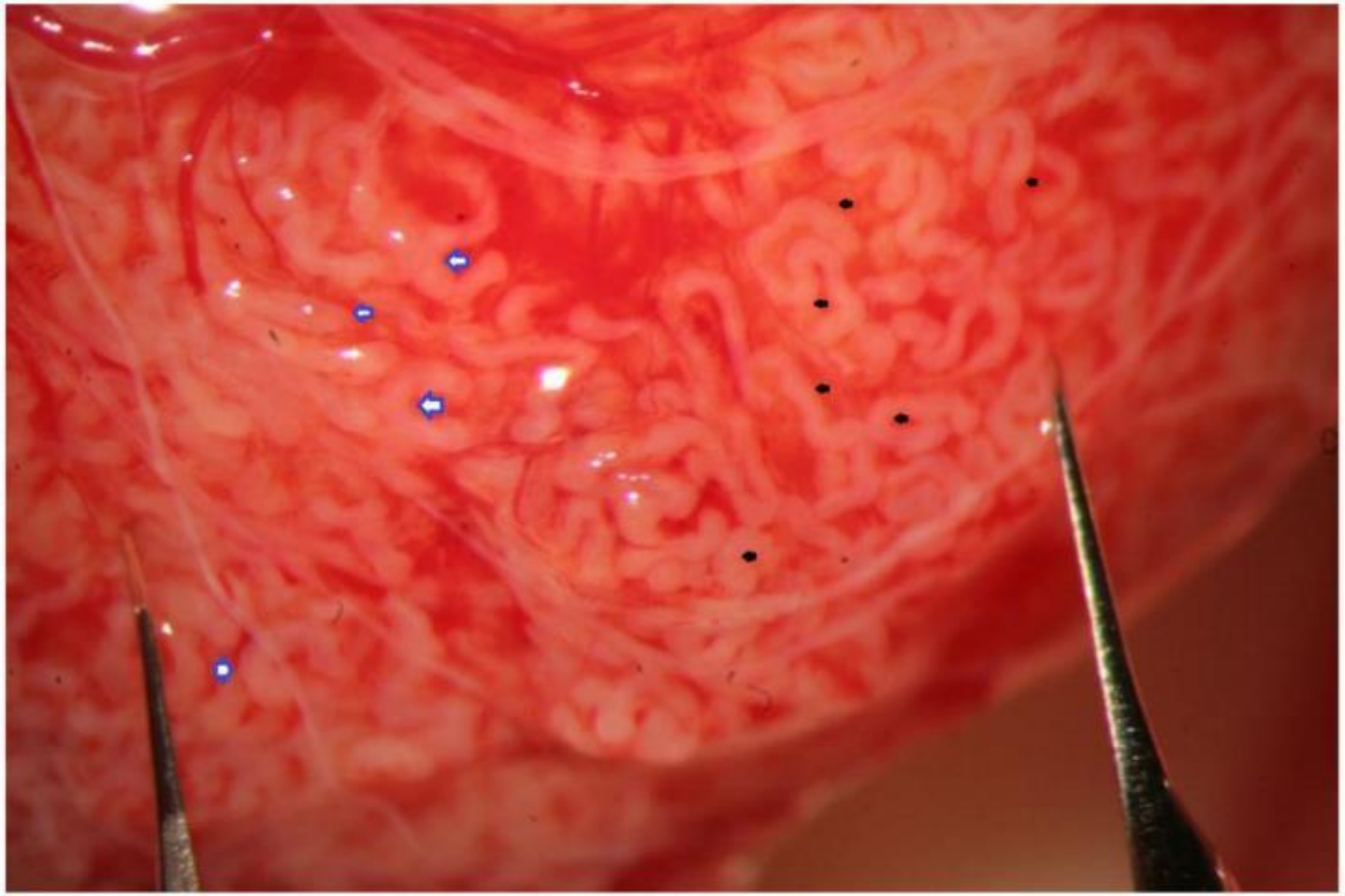
Bouker et al. 2019. *Basic and Clinical Andrology* 29 (July): 11.

Healthy seminiferous tubulues



Colpi et al. 2021. *Journal of Clinical Medicine Research* 10 (19).

Healthy seminiferous tubulues



Colpi et al. 2021. *Journal of Clinical Medicine Research* 10 (19).

Testicle with diffuse maturation arrest



Bouker et al. 2019. *Basic and Clinical Andrology* 29 (July): 11.

Pocket of healthy seminiferous tubules

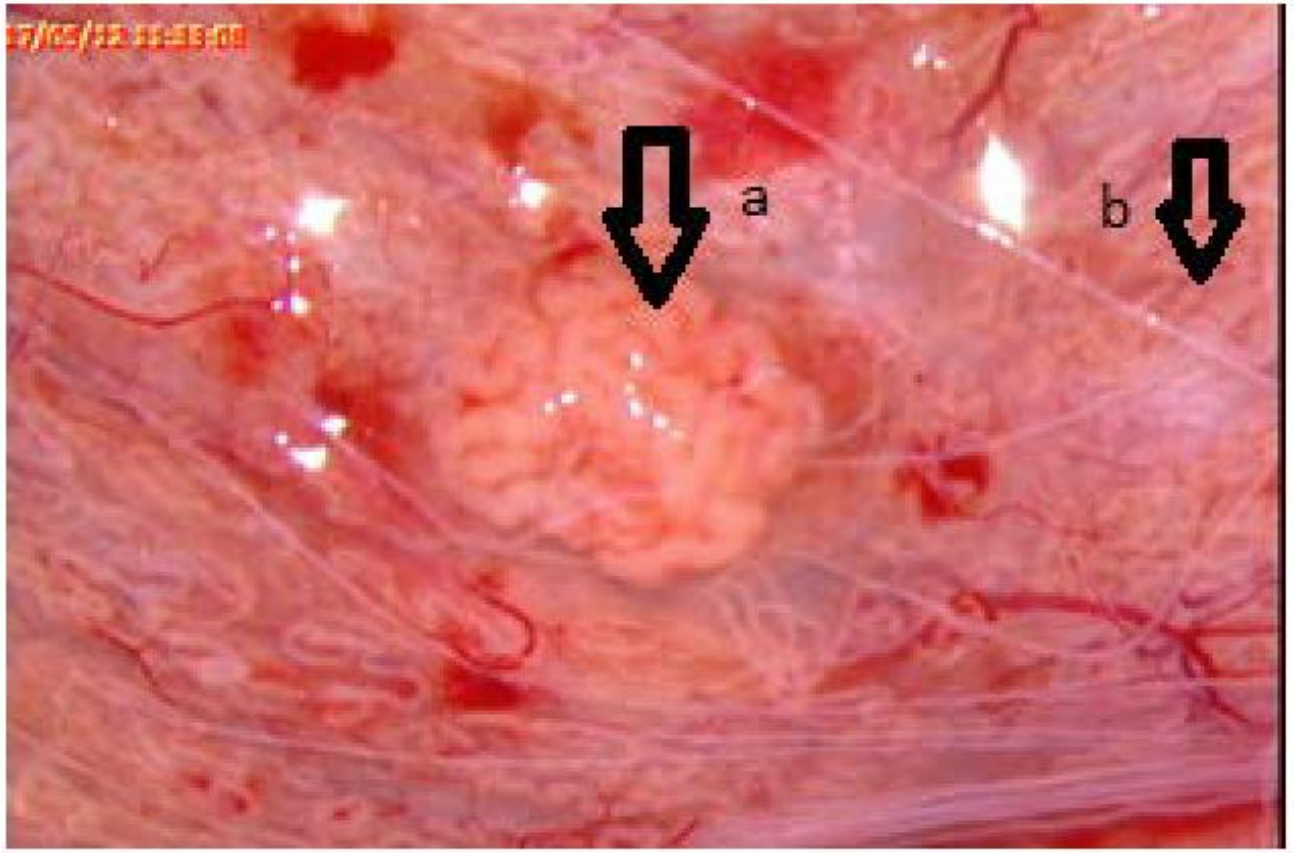
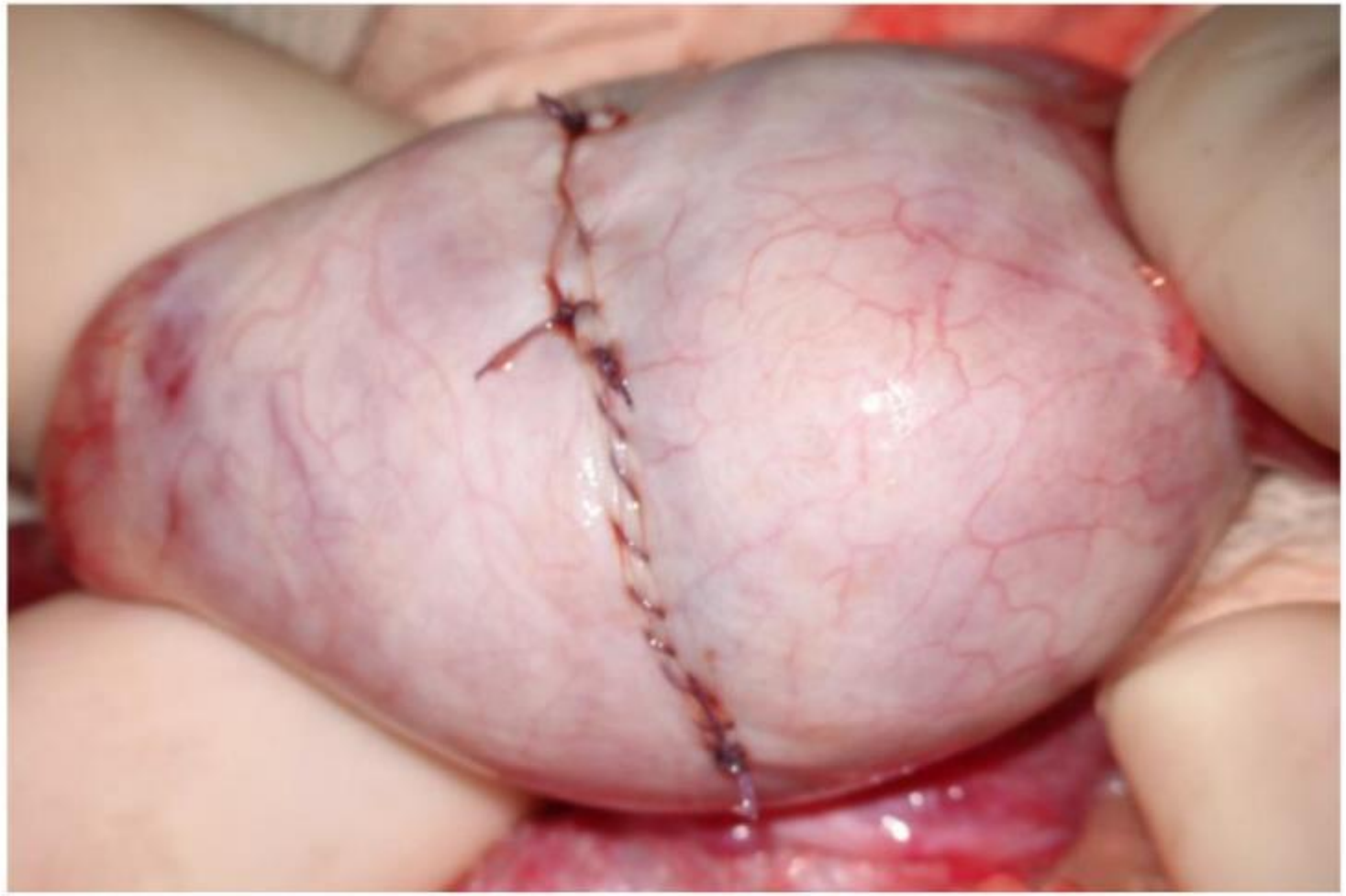


Figure 1: Testicular tissue with aid of operating microscope at x25

Sankalp and Dharmaraj. 2014. *Andrology* 03 (01).

Closure of testicle



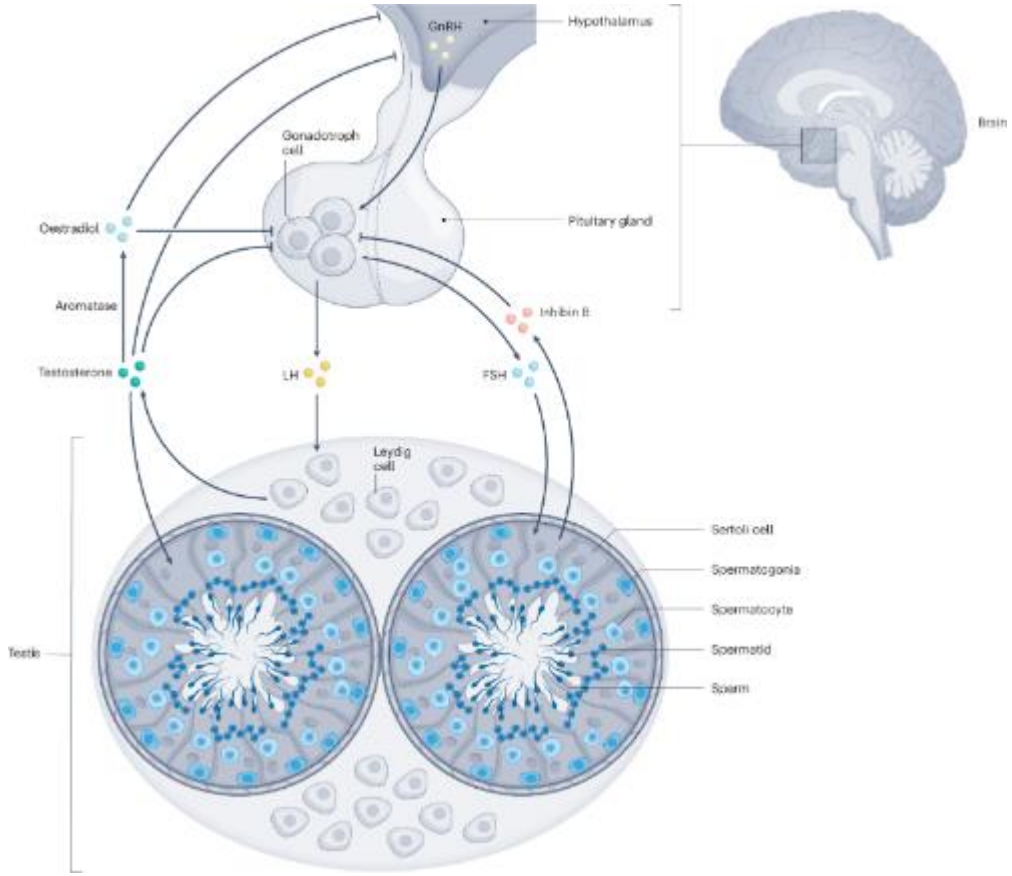
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KS: Fertility management

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- If no sperm in ejaculate (92%) → **microsurgical testicular sperm extraction (microTESE)** → intracytoplasmic sperm injection (ICSI)
  - **~50%** sperm retrieval rate (SRR)
- Only **10%** of KS couples go on to have biological child
  - If unsuccessful → repeat microTESE, donor sperm, adoption

Hypogonadotropic hypogonadism

- Anosmic (Kallman's syndrome) versus non-anosmic
- **Congenital failure of GnRH secretion** by the hypothalamus → deficient FSH, LH and testosterone levels (hypogonadotropic hypogonadism) → impaired spermatogenesis
- Treatment
  - If no desire for fertility → testosterone replacement therapy (TRT)
  - If desire for fertility → gonadotropin replacement therapy to promote intratesticular testosterone (ITT) production and spermatogenesis
    - Recombinant FSH
    - Human chorionic gonadotropin (hCG)
      - Analogue of LH



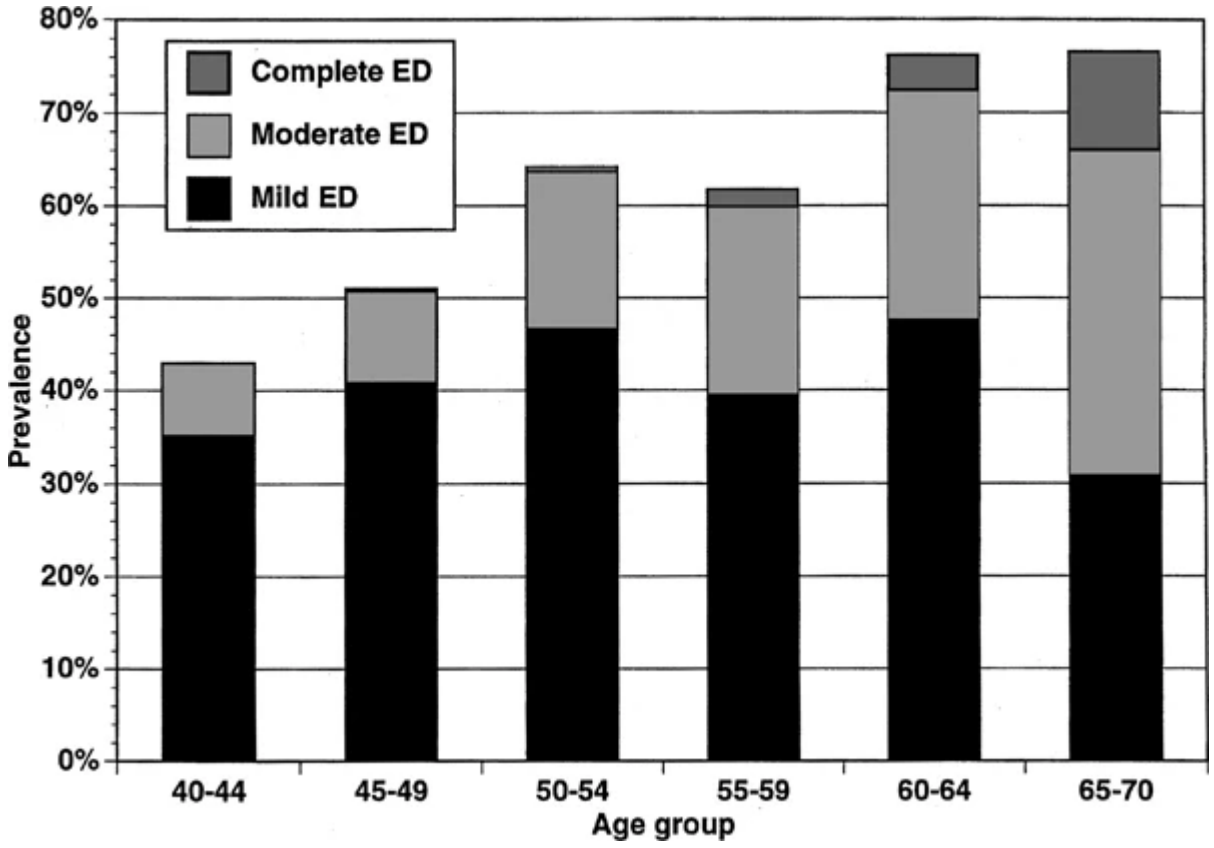
Eisenberg et al. Male infertility. *Nature Reviews Disease Primers*. 2023;9:49

Retrograde ejaculation

- Semen goes into the bladder instead of out through the urethra (**dry orgasm**)
- Causes
  - Surgeries that disrupt the bladder neck or sympathetic innervation
    - **Transurethral resection of the prostate (TURP)**
    - Retroperitoneal lymph node dissection (RPLND)
  - Medications that relax the bladder neck
    - Most common: Alpha-adrenergic antagonists (e.g. **tamsulosin**)
  - Neuropathies
    - **Diabetic autonomic neuropathy**
    - Spinal cord injury
    - Other neurologic disorders (e.g. multiple sclerosis, Parkinson's disease)
- Diagnosis: Post-ejaculatory urinalysis
  - Presence of significant number of sperm in urine is diagnostic
- Treatment
  - Medical therapy
    - Alpha-adrenergic antagonists (e.g. phenylephrine)
    - Tricyclic antidepressants (e.g. imipramine)
  - Sperm collection from urine → intrauterine insemination (IUI), assisted reproductive techniques (ART)

Erectile dysfunction (ED)

- Inability to attain and/or maintain penile erection sufficient for satisfactory sexual performance
- Affects up to 50 million American men
- Prevalence generally matches patient age
  - 40% of 40-year-old men
  - 60% of 60-year-old men
  - 80% of 80-year-old men



https://www.nature.com/articles/3901010/figures/1

ED: Types

- Vasculogenic
- Neurogenic
- Psychogenic
- Hormonal
- Drug-induced

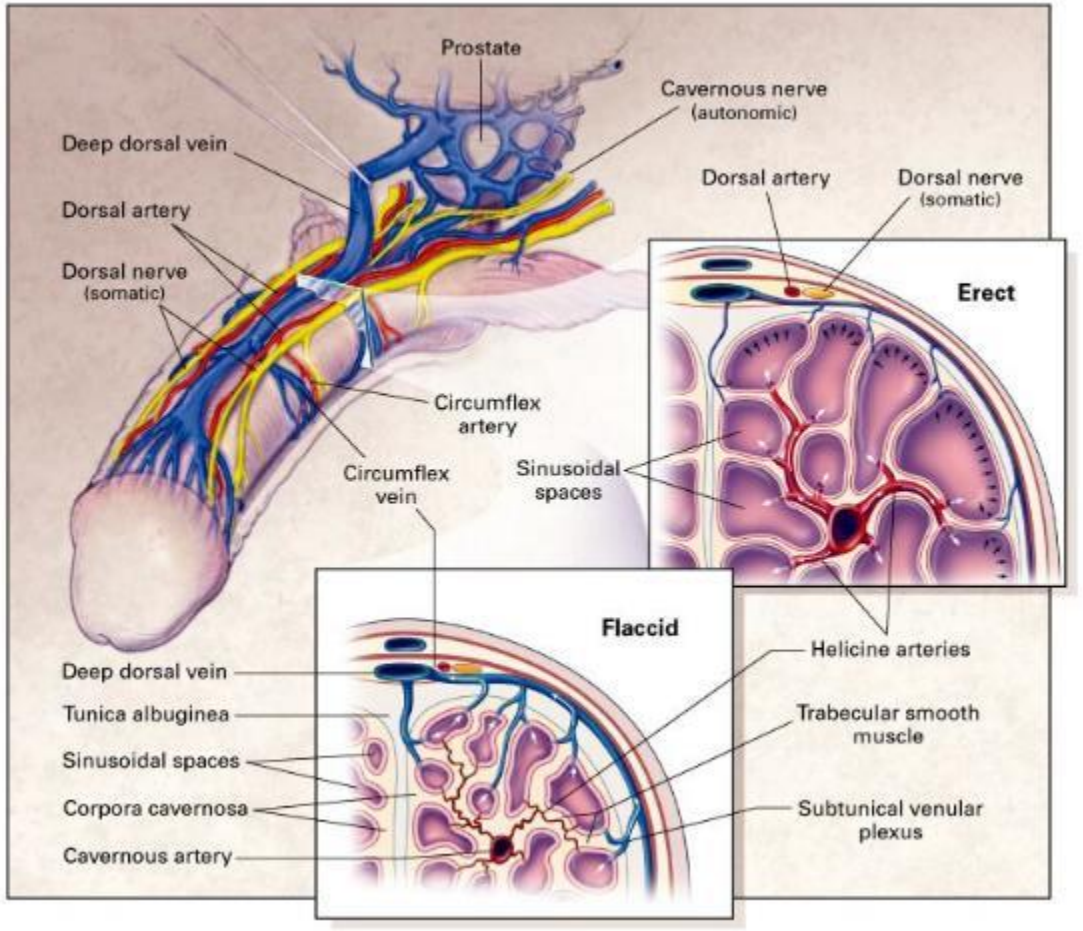
TABLE 1. CLASSIFICATION AND COMMON CAUSES OF ERECTILE DYSFUNCTION.

CATEGORY OF ERECTILE DYSFUNCTION	COMMON DISORDERS	PATHOPHYSIOLOGY
Psychogenic	Performance anxiety Relationship problems Psychological stress Depression	Loss of libido, overinhibition, or impaired nitric oxide release
Neurogenic	Stroke or Alzheimer's disease Spinal cord injury Radical pelvic surgery Diabetic neuropathy Pelvic injury	Failure to initiate nerve impulse or interrupted neural transmission
Hormonal	Hypogonadism Hyperprolactinemia	Loss of libido and inadequate nitric oxide release
Vasculogenic (arterial or cavernosal)	Atherosclerosis Hypertension Diabetes mellitus Trauma Peyronie's disease	Inadequate arterial flow or impaired veno-occlusion
Drug-induced	Antihypertensive and antidepressant drugs Antiandrogens Alcohol abuse Cigarette smoking	Central suppression Decreased libido Alcoholic neuropathy Vascular insufficiency
Caused by other systemic diseases and aging	Old age Diabetes mellitus Chronic renal failure Coronary heart disease	Usually multifactorial, resulting in neural and vascular dysfunction

https://www.nejm.org/doi/full/10.1056/NEJM200006153422407

Vasculogenic ED

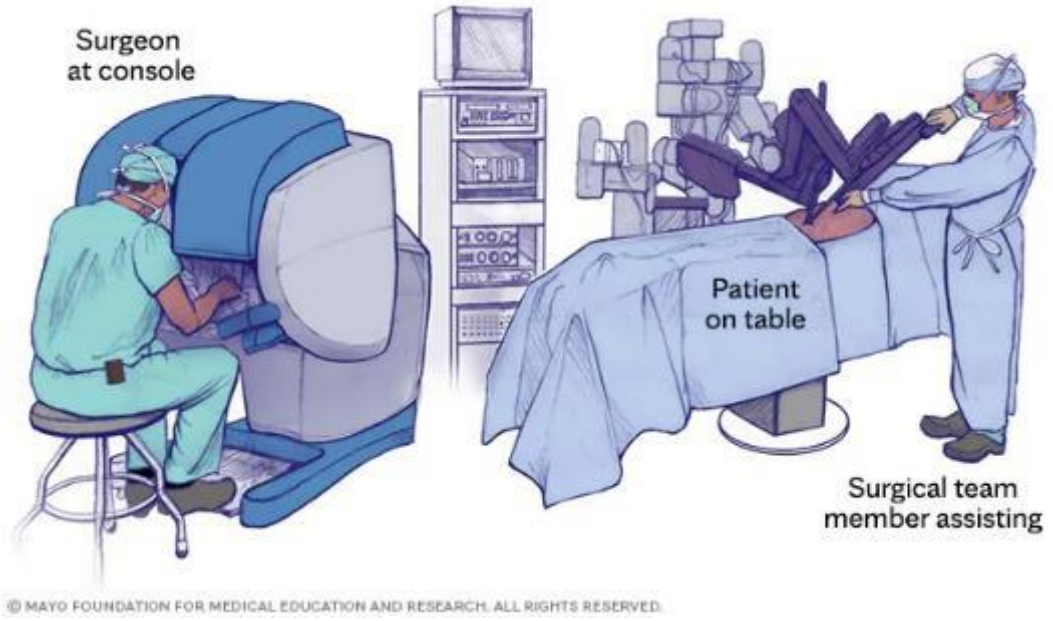
- Most prevalent form, and the most common cause in men aged >50 years
- Due to impaired arterial inflow into the penis (**arterial insufficiency**), failure of veno-occlusive mechanism (**venous leak**), or both
- Associated with cardiovascular disease, diabetes mellitus, hypertension, hyperlipidemia, and nicotine use
- Considered to be a **harbinger of future cardiac events**
    - Men with new-onset vasculogenic ED should undergo cardiac evaluation



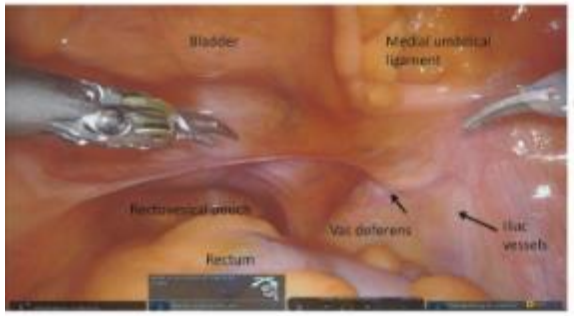
https://www.nejm.org/doi/full/10.1056/NEJM200006153422407

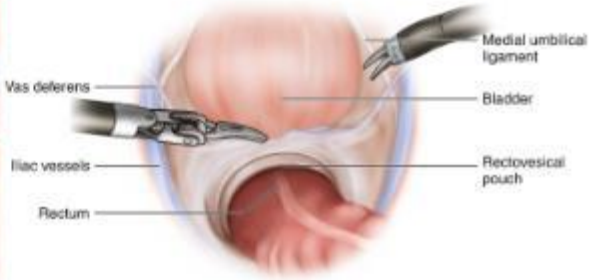
Neurogenic ED

- Due to disruption of neural pathways involved in erection
- Examples of causes
  - Stroke, spinal cord injury, pelvic injury or surgery (e.g. **radical prostatectomy**, abdominoperineal resection), diabetic neuropathy
- Older men who undergo radical prostatectomy for prostate cancer often end up with severe ED due to combination of vasculogenic and neurogenic causes



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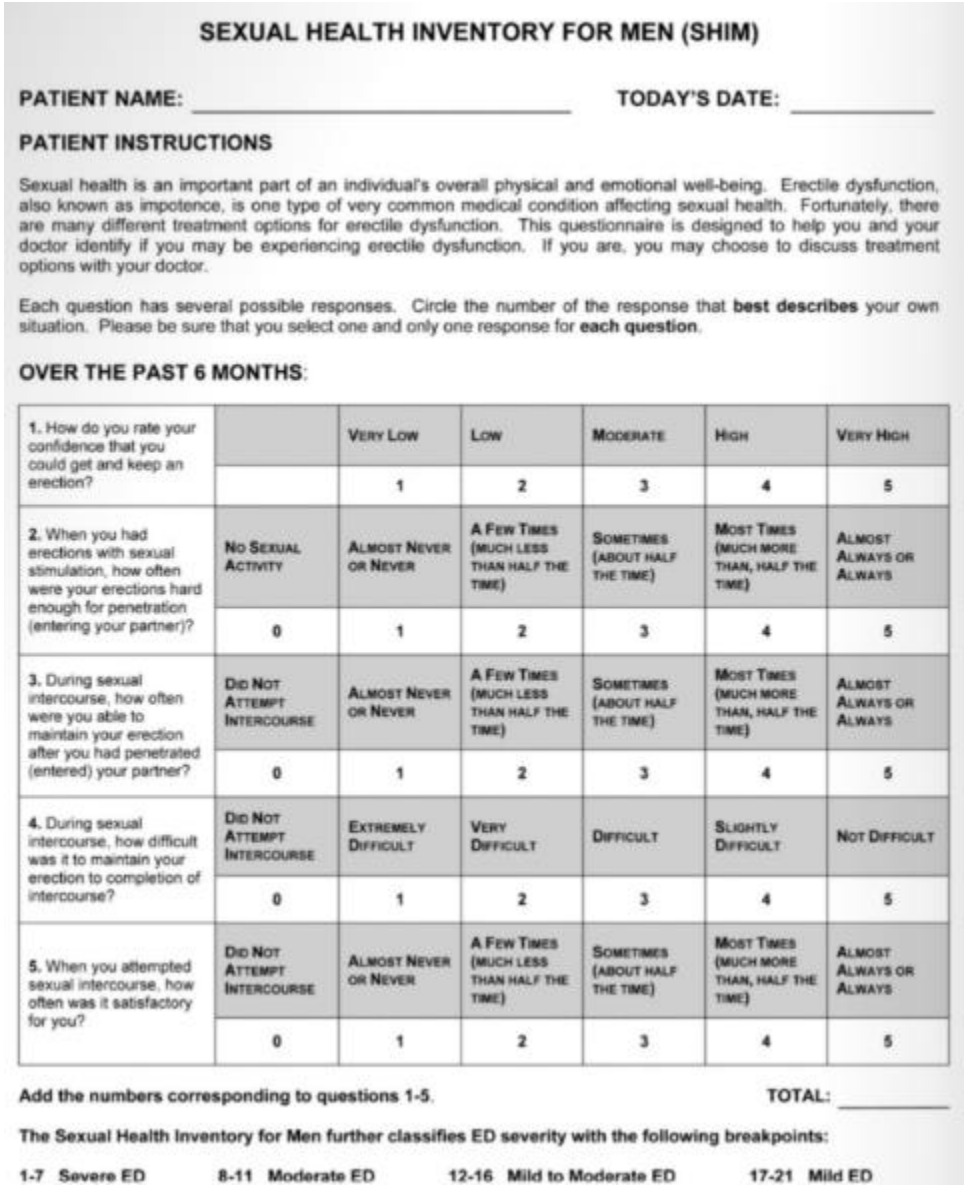


Psychogenic ED

- More common in younger men
- Characterized by normal penile blood flow and nerve function but impaired erection due to psychological factors
- Common causes
  - Performance anxiety, depression, relationship conflict, stress, psychiatric disorders
- If patient has issues with partnered sexual activity but has no issues achieving and maintaining an erection with masturbation, it is usually a sign of psychogenic ED
  - These patients also tend to have **spontaneous nocturnal/morning erections**

ED: Clinic evaluation

- Screening questionnaire
  - International Index of Erectile Function (IIEF)
  - Sexual Health Inventory for Men (SHIM)
- Focused medical and surgical history
- Medication history
- Psychological and relationship assessment
- Physical exam to assess obesity, neuro-vascular stigmata and signs of hypogonadism



SEXUAL HEALTH INVENTORY FOR MEN (SHIM)

PATIENT NAME: _____ **TODAY'S DATE:** _____

PATIENT INSTRUCTIONS

Sexual health is an important part of an individual's overall physical and emotional well-being. Erectile dysfunction, also known as impotence, is one type of very common medical condition affecting sexual health. Fortunately, there are many different treatment options for erectile dysfunction. This questionnaire is designed to help you and your doctor identify if you may be experiencing erectile dysfunction. If you are, you may choose to discuss treatment options with your doctor.

Each question has several possible responses. Circle the number of the response that **best describes** your own situation. Please be sure that you select one and only one response for **each question**.

OVER THE PAST 6 MONTHS:

1. How do you rate your confidence that you could get and keep an erection?		Very Low	Low	Moderate	High	Very High
		1	2	3	4	5
2. When you had erections with sexual stimulation, how often were your erections hard enough for penetration (entering your partner)?	No Sexual Activity	Almost Never or Never	A Few Times (much less than half the time)	Sometimes (about half the time)	Most Times (much more than, half the time)	Almost Always or Always
	0	1	2	3	4	5
3. During sexual intercourse, how often were you able to maintain your erection after you had penetrated (entered) your partner?	Did Not Attempt Intercourse	Almost Never or Never	A Few Times (much less than half the time)	Sometimes (about half the time)	Most Times (much more than, half the time)	Almost Always or Always
	0	1	2	3	4	5
4. During sexual intercourse, how difficult was it to maintain your erection to completion of intercourse?	Did Not Attempt Intercourse	Extremely Difficult	Very Difficult	Difficult	Slightly Difficult	Not Difficult
	0	1	2	3	4	5
5. When you attempted sexual intercourse, how often was it satisfactory for you?	Did Not Attempt Intercourse	Almost Never or Never	A Few Times (much less than half the time)	Sometimes (about half the time)	Most Times (much more than, half the time)	Almost Always or Always
	0	1	2	3	4	5

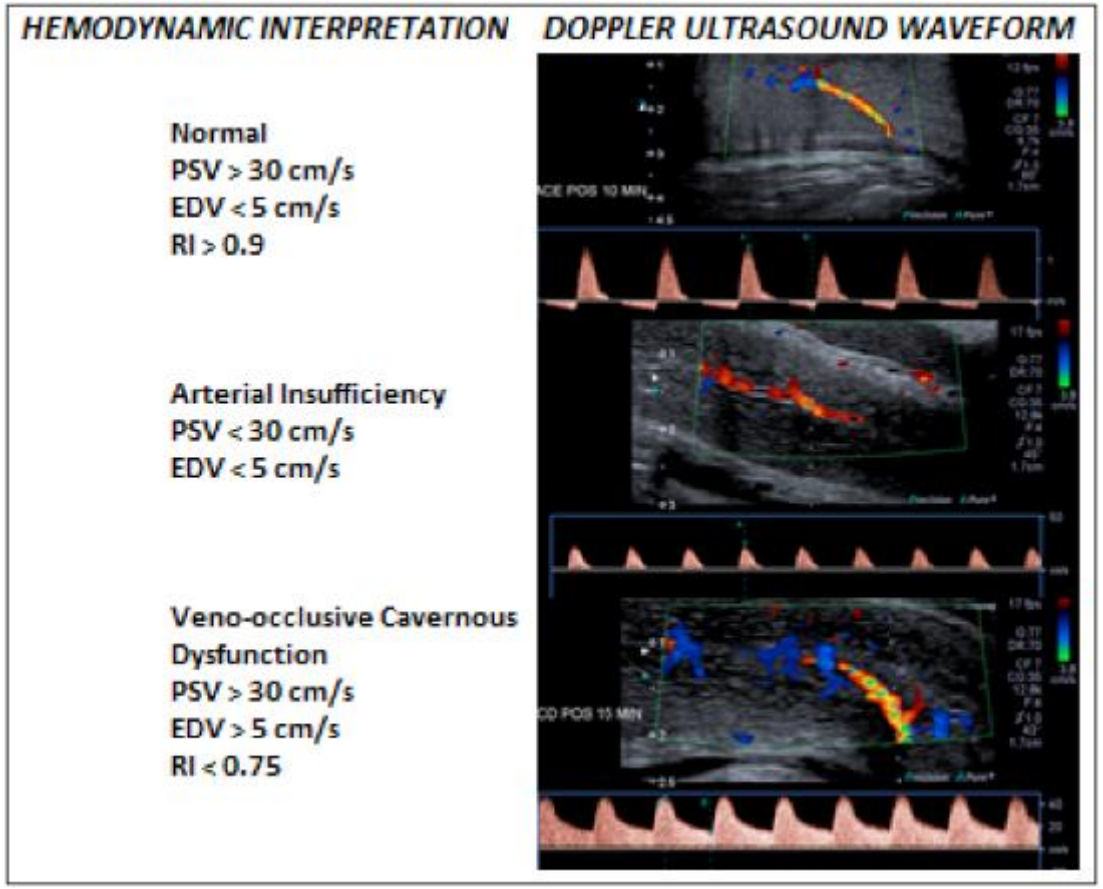
Add the numbers corresponding to questions 1-5. **TOTAL:** _____

The Sexual Health Inventory for Men further classifies ED severity with the following breakpoints:

1-7 Severe ED **8-11 Moderate ED** **12-16 Mild to Moderate ED** **17-21 Mild ED**

ED: Additional evaluation

- Serum testosterone, glucose/hemoglobin A1c, and serum lipids may be checked to assess for etiologies
- Duplex Doppler ultrasound with intracavernosal vasodilator injection
  - A medication that induces erection (e.g. Trimix) is injected into the corpus cavernosum
  - Once erection is achieved, Duplex Doppler ultrasound is used to assess the **peak systolic velocity (PSV)** and **end-diastolic velocity (EDV)** of the cavernosal arteries
    - **PSV <25 cm/sec: Arterial insufficiency**
    - **EDV >5 cm/sec: Venous leak**
  - Not commonly performed

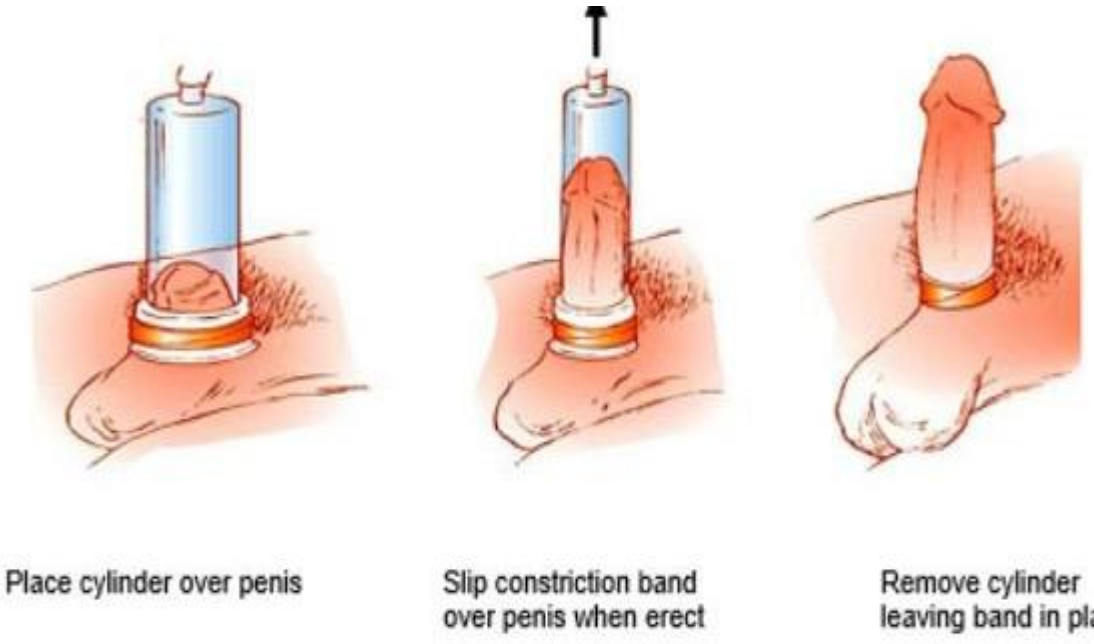


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ED: Treatments

- Behavioral modification (e.g. weight loss, physical activity, nicotine cessation, stress reduction)
- Sex therapy / couples therapy
- Vacuum erection device (VED)
  - Can be used together with penile constriction band (“cock ring”) to help maintain the erection
- Medication adjustment (e.g. changing class of antihypertensive or antidepressant)





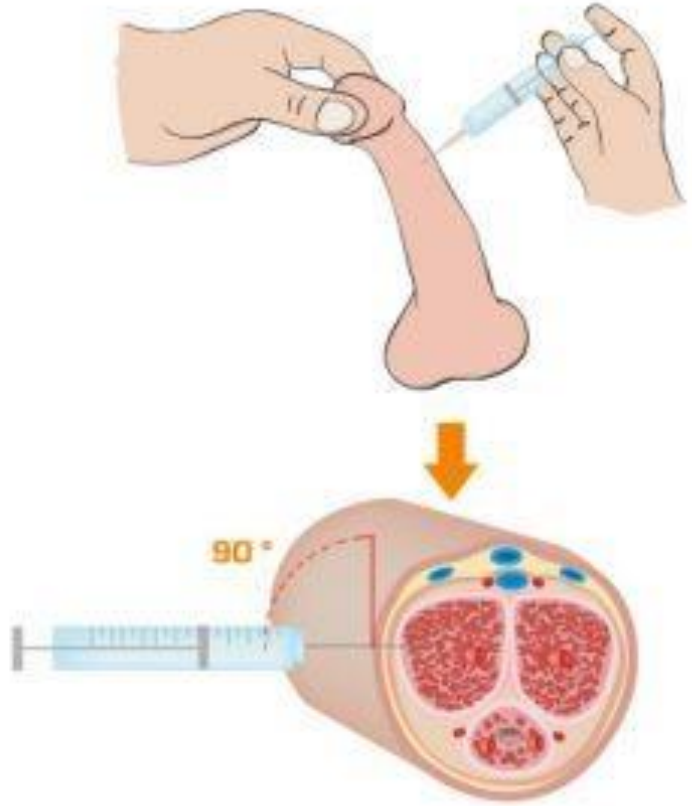
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Oral phosphodiesterase-5 (PDE5) inhibitors

- Sildenafil, tadalafil, avanafil, vardenafil
- Mechanism of action
  - Inhibition of PDE5 → **increased penile cavernosal cGMP** → smooth muscle relaxation in corpus cavernosum vasculature → increased erection hardness and duration
- Usually taken as needed prior to sexual activity
  - Tadalafil has a long half-life (17.5 h) and can be taken daily at low dose
- Common side effects
  - Headache, flushing, nasal congestion, myalgia
- **Contraindicated in patients who have nitrate prescriptions** (e.g. nitroglycerin for chest pain)
  - Concomitant use of PDE5 inhibitor and nitrate can cause severe hypotension

Intracavernosal injection (ICI)

- Direct injection of vasodilators into the penile corpus cavernosum → direct smooth muscle relaxation
- Safe to prescribe to patients who use nitrates
- Commonly used ICI formulations
  - Alprostadil (synthetic prostaglandin E1)
    - Efficacy: ~70% for men who failed oral PDE5 inhibitors
  - Bimix (papaverine + phentolamine)
    - Limited efficacy data
  - **Trimix (alprostadil + papaverine + phentolamine)**
    - Efficacy: Up to 90% in men who failed oral PDE5 inhibitors



Ischemic priapism

- Most feared side effect of ICI (1 – 4% risk)
- Painful erection persisting **≥4 hours,** characterized by rigid shaft but **soft glans**
- Urologic emergency that necessitates prompt evaluation in ER
  - Priapism persisting **>36 hours** → permanent, complete ED as well as significant and irreversible penile scarring and shortening
- Diagnosis: ABG performed on cavernosal blood
  - **PO_2 <30 mmHg, PCO_2 >60 mmHg, pH <7.25**
- Initial treatment: Bedside penile irrigation and aspiration with normal saline as well as **phenylephrine** injections
  - Phenylephrine directly stimulates alpha-1 adrenergic receptors in the smooth muscle of the corpora cavernosa → vasoconstriction, decreased arterial inflow, and increased venous outflow → detumescence
- Refractory cases require surgical intervention (“shunt”)



Penile implant

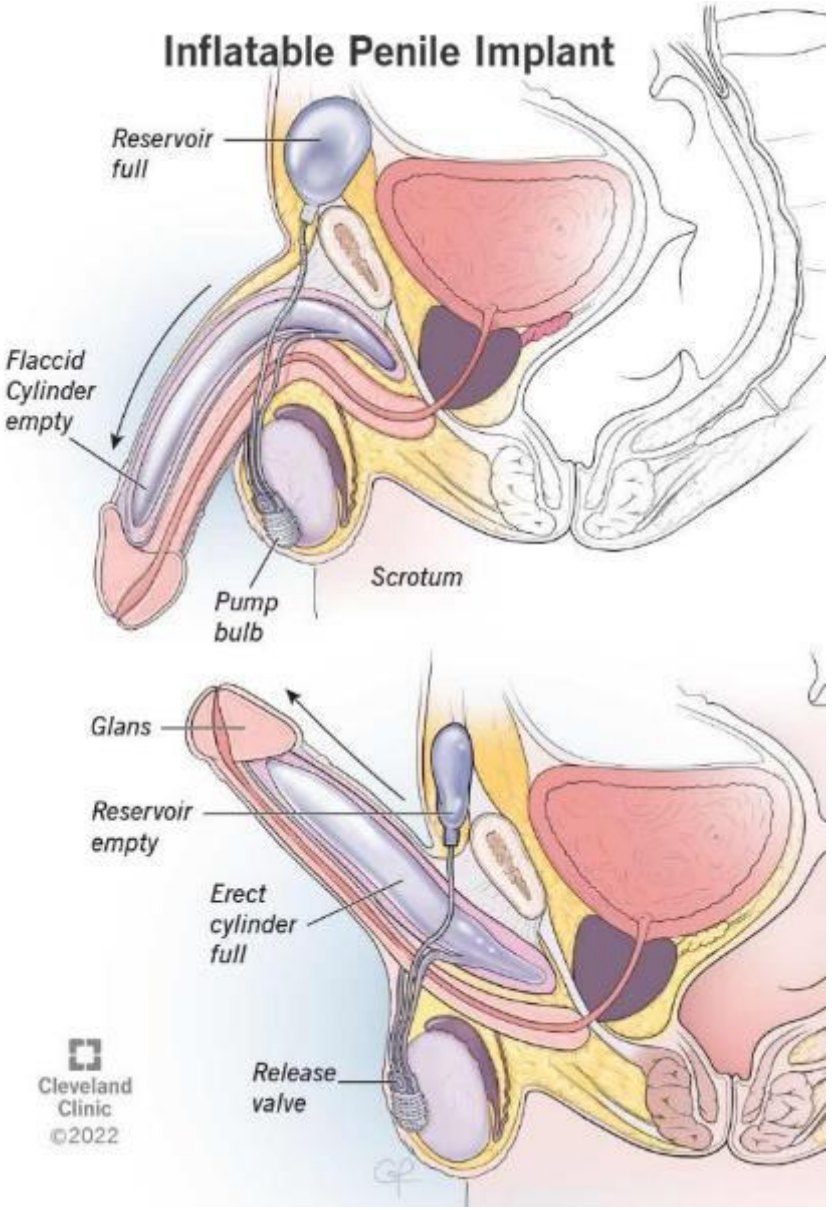
- 2 major types
  - Inflatable penile prosthesis (IPP)
  - Malleable penile prosthesis (MPP)





Penile implant

- 2 major types
  - Inflatable penile prosthesis (IPP)
  - Malleable penile prosthesis (MPP)
- Essentially a cure for ED and allows for spontaneous penetrative intercourse
- Irreversible procedure
- **Highest satisfaction among all ED treatments** (>95% in patients and partners)
- Infection of the implant occurs in 1 – 4% of first-time cases and 10% of revision/replacement cases
  - Requires explantation of all implant components in addition to antibiotic therapy



Thank you